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MQuant: Unleashing the Inference Potential of Multimodal Large Language Models via Static Quantization

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Abstract

Multimodal large language models (MLLMs) have garnered widespread attention due to their ability to understand multimodal input. However, their large parameter sizes and substantial computational demands severely hinder their practical deployment and application. While quantization is an effective way to reduce model size and inference latency, its application to MLLMs remains underexplored. In this paper, we propose MQuant, a post-training quantization (PTQ) framework designed to tackle the unique challenges of multimodal large language models (MLLMs). Conventional quantization often struggles with MLLMs because of (a) high inference latency from large visual token counts, (b) distributional disparities between visual and textual tokens, and (c) extreme outliers introduced by Hadamard-based transformations. To address these issues, MQuant introduces: • Modality-Specific Static Quantization (MSQ), assigning distinct static scales for visual vs. textual tokens; • Attention-Invariant Flexible Switching (AIFS), re-ordering tokens to preserve casual attention while eliminating expensive token-wise scale computations; • Rotation Magnitude

Suppression (RMS), mitigating weight outliers arising from on-line Hadamard rotations. On five mainstream MLLMs (including Qwen-VL, MiniCPM-V, CogVLM2), MQuant under W4A8 achieves near-floating-point accuracy (<1% degradation) while reducing inference latency by up to 30%, significantly outperforming existing PTQ baselines. Our MQuant effectively bridges the gap for efficient and accurate MLLMs inference in resource-constrained devices. Code will be released in <https://github.com/StiphyJay/MQuant>.

CCS Concepts

• Computing methodologies → Natural language generation; Visual content-based indexing and retrieval.

Keywords

Multimodal Large Language Model, Model Compression, Model Efficiency, Post-Training Quantization

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1 Introduction

Recent advances in large language models (LLMs) [7, 16, 39, 40] have led to remarkable performance on a wide range of natural language processing tasks. However, these models often struggle when dealing with non-textual data such as images or videos. Multimodal large language models (MLLMs) [1, 33, 45] address this limitation

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by integrating visual and textual modalities, thereby enabling comprehensive understanding and reasoning. Despite these benefits, their large parameter sizes, coupled with substantial computational demands, poses a major challenge for real-world deployment, particularly in resource-constrained or privacy-sensitive environments.

Challenges in MLLMs Quantization. Quantization has proven an effective strategy for reducing memory usage and inference costs in LLMs [35, 52, 59], by converting high-precision parameters (e.g., FP32) into lower-bit representations (e.g., INT8). Yet, the transition from LLMs to MLLMs brings three unique difficulties: ❶ **Time-to-First-Token (TTFT) Explosion.** MLLMs often generate large numbers of visual tokens (e.g., patch embeddings or region proposals) with the resolution and aspect ratio of input images or videos. As shown in Fig 1(a), in models such as Qwen2-VL [44], the number of prefill visual tokens grows as image resolution increases (detailed figure in Appendix A.6). As image or video resolution increases, the prefill visual tokens can escalate dramatically—leading to high TTFT and negatively impacting latency-sensitive tasks. Moreover, in higher-demand scenarios such as video-based tasks and multi-image dialogues, the accumulation of visual tokens becomes even more pronounced, further exacerbating TTFT growth. Although the commonly adopted per-token dynamic quantization [6, 12, 30, 44] offers flexibility to accommodate the larger activation distribution variance across different tokens in vision modal, its requirement for online token-wise scale recalculation during inference significantly amplifies this computational overhead. Moreover, recent studies [9, 37] have underscored the inefficiencies of per-token dynamic quantization, especially for resource-constrained edge devices, where such methods exacerbate latency and memory pressures. ❷ **Disparate Modality Distributions.** As shown in Fig 1 (b), the activation distributions between visual and textual tokens reveal substantial numerical discrepancies, where visual token activations can span a significantly broader range (e.g., -20 to 10) than textual tokens, which typically center near 0 . A single global scale factor for both visual and textual tokens leads to either aggressive clipping of visual outliers or increased quantization granularity for text, harming overall accuracy. ❸ **Visual Token Outliers.** High-magnitude outliers in visual tokens, often reflecting salient image regions, make traditional clipping-based methods unsuitable. As shown in Table 1, naive clipping outliers significantly degrades accuracy, highlighting the necessity for judicious outlier treatment.

Table 1: Ablations on the clip range for different tokens.

Clip Range	Bits (W/A)	Visual Tokens	Textual Tokens
-	BF16 / BF16	61.40	
(0-1.0)	BF16 / INT16	61.20 ($\downarrow 0.20$)	61.25 ($\downarrow 0.15$)
(0-0.99999)	BF16 / INT16	18.92 ($\downarrow 42.48$)	60.09 ($\downarrow 1.31$)

To this end, we propose *MQuant*, an accurate and efficient post-training quantization (PTQ) solution specifically designed for multimodal large language models (MLLMs). First, to reduce the TTFT while maintaining accuracy, we introduce **Modality-Specific Static Quantization (MSQ)**, which apply distinct static per-tensor scales for visual vs. textual tokens based on the distinct distribution differences. Second, for further acceleration, we design an **Attention-Invariant Flexible Switching (AIFS)** scheme. AIFS transforms mixed multimodal tokens into unified, modality-decoupled tokens

without the need for dynamic position vectors, maintaining computational equivalence and memory efficiency by avoiding the overhead associated with dynamic processing. Based on MSQ and AIFS, *MQuant* slashes TTFT and preserves accuracy. Third, we reveal the weight outliers caused by the online Hadamard rotations through theoretical analysis and propose **Rotation Magnitude Suppression (RMS)** to mitigate them. Our theoretical analysis reveals the emergence of large mean channels in MLLM weights, and RMS effectively reduces these outliers with minimal overhead.

We evaluate *MQuant* on five mainstream MLLMs, including InternVL [11], Qwen-VL [6], MiniCPM-V [50], CogVLM2 [19] and Qwen2-VL [44]. The extensive results on diverse multimodal reasoning tasks demonstrate that *MQuant* achieves less than 1% accuracy loss on all MLLMs with 23% and 100% speedup in prefill and decode stage under the W4A8 setting, highlighting its superior performance and wide practical value. Our main contributions are summarized as follows:

- **Insightful Observation.** To our best knowledge, we present the first comprehensive analysis of quantization issues in MLLMs, revealing the root causes of performance collapse and identifying inference speed bottlenecks as well as the quantization challenges posed by modality differences.
- **MSQ and AIFS.** We design Modality-specific Static Quantization (MSQ) and Attention-Invariant Flexible Switching (AIFS) to accelerate inference for heterogeneous variable-sequence multimodal inputs while maintaining accuracy.
- **RMS.** We identify weight outliers caused by online rotation, and propose Rotation Magnitude Suppression to enhance quantization performance.
- **MQuant.** We propose *MQuant*, a general PTQ framework designed for MLLMs, demonstrating both near-lossless performance and with significant speedup.

2 Preliminaries and Related Work

2.1 Multimodal Large Language Models

The existing MLLMs framework [6, 12] mainly consists of three modules: (1) **Visual encoder E** for processing visual inputs, (2) **Vision-language projector P** for aligning the text and visual modalities, (3) **Large language model LLM** that handles the multimodal tokens and performs reasoning. MLLMs' foundational components have greatly benefited from rapid advancements in LLM technology. Flamingo [3] pioneered connecting pre-trained visual encoders to LLMs, demonstrating strong generalization across visual-language tasks. Following ChatGPT [2], numerous open-source models based on pre-trained LLMs, such as LLaMA series [16, 39, 40], have been proposed [22, 24, 30, 60]. Subsequent efforts, including Qwen-VL [6], InternVL [12], and CogVLMV2 [19], have enhanced MLLMs by improving high-resolution inputs and scaling training data. Besides, smaller MLLMs like Mini-Gemini [28], MobileVLM [13], and MiniCPM-V [50] have emerged. Despite those advancements, the large parameter sizes of MLLMs lead to high computational costs, yet dedicated quantization methods to reduce memory usage and accelerate inference still remain underexplored.

2.2 PTQ for LLMs/MLLMs

PTQ [23, 51, 58] serves as a potential strategy for model compression. By converting the high-precision variables of pre-trained

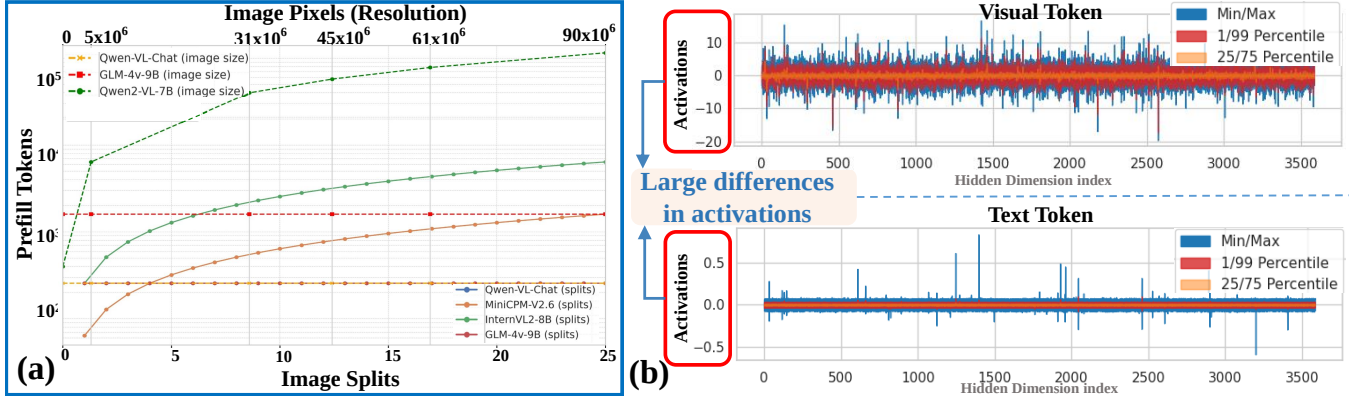


Figure 1: (a) Prefill visual tokens counts across different MLLMs as the image splits/resolution increases. (b) The activation values of visual tokens range from -20 to 10 , while textual tokens concentrate near 0 (rarely exceeding ± 0.5).

models into low-bit integers, it achieves memory reduction and inference speed acceleration. For uniform quantization, given a floating-point (FP) tensor x (weights or activations), it can be uniformly quantized to b -bits in signed quantization as follows:

$$\hat{x} = Q_U(x, b) = (\text{clamp}(\lfloor \frac{x}{s} \rfloor + z, q_{\min}, q_{\max}) - z) \cdot s \quad (1)$$

where $s = \frac{\max(|x|)}{2^{b-1}-1}$ is the scale factor, $\lfloor \cdot \rfloor$ is the rounding-to-nearest operator, and the function $\text{clamp}(\cdot)$ clips values outside the integer range $[q_{\min}, q_{\max}]$. z is zero-point. s denotes the quantization scale factor, which reflects the proportional relationship between FP values and integers. $[q_{\min}, q_{\max}]$ is the quantization range determined by the bit-width b . Generally, when we quantize the network's weight with 4-bit and activations with 8-bit, called it as W4A8. We can calculate s offline using the activations from calibration samples, known as **static quantization**. We can also use the runtime statistics of activations to get s , referred to as **dynamic quantization**. More details are in Appendix A. 12.

Existing post-training quantization (PTQ) methods for LLMs are categorized into weight-only and weight-activation quantization [10, 20, 21, 25, 26, 34, 48, 49, 54, 55, 57]. Weight-only methods like GPTQ [17], QuIP [8], and AWQ [29] achieve high compression rates but offer limited inference acceleration. In contrast, weight-activation quantization methods [46, 47, 53, 56] quantize both weights and activations, improving memory usage and latency. The main challenge is activation outliers causing quantization errors. Techniques like SmoothQuant [47] shift quantization difficulty from activations to weights, while OmniQuant [35] optimizes performance by training quantization parameters. SliceGPT [4] reduces memory demands by designing a Pre-LN + Rotate Scheme for LLMs sparsification based on computational invariance. Recently, Quarot [5] introduces applies *offline* Hadamard transforms and a *partially online* Hadamard transform to eliminate outliers, achieving state-of-the-art quantization results on LLMs. However, this solution is not applicable to MLLMs due to inherent modality differences. Building on the above studies, research on PTQ specifically tailored for MLLMs remains notably scarce. Existing studies, such as Q-VLM [43], propose a block-wise quantization framework that optimizes cross-layer dependencies via entropy-guided partitioning. MBQ [27] uses gradients of the supervised fine-tuning loss

with respect to vision and language tokens as sensitivity indicators to balance reconstruction loss during calibration. Despite efforts, existing methods still employ per-token dynamic quantization for activation which inherently introduces significant computational overhead than per-tensor static quantization discussed earlier. In contrast, our MQuant framework addresses these challenges by extending full static quantization to both weights and activations, specifically designed to unleash the inference potential of MLLMs while overcoming the modality-specific limitations faced by LLM-focused PTQ methods, which are not directly applicable to MLLMs due to significant modality differences.

3 Method

In this section, we present *MQuant*, a post-training quantization solution specifically designed for MLLMs. In Sec. 3.1, we describe proposed modality-specific static quantization (MSQ) and attention-invariant flexible switching (AIFS) mechanisms. In Sec. 3.2, we identify the weight outliers caused by the online Hadamard rotations and state Rotation Magnitude Suppression (OMS).

3.1 Modality-Specific Static Quantization and Attention-Invariant Flexible Switching

We address this issue with a two-part method: (1) **Modality-Specific Static Quantization (MSQ)** and (2) **Attention-Invariant Flexible Switching (AIFS)**. MSQ applies different static scaling factors for visual and textual tokens, while AIFS reorders their positions to avoid irregular data slicing. MSQ+AIFS retains accuracy while eliminating the expensive token-wise scale online computation during inference. This approach achieves efficiency gains by leveraging per-tensor static quantization, especially for high-resolution inputs.

Modality-Specific Static Quantization (MSQ). Let E be an input sequence of length L that intermixes textual and visual tokens. Denote $E = \{e_1^t, \dots, e_{m-1}^t, e_m^v, \dots, e_n^v, e_{n+1}^t, \dots, e_L^t\}$, where m, n specify the visual token segment. We observe that visual tokens often have larger activation magnitudes, which can overshadow textual features. To handle these differences, we apply two distinct sets of *static per-tensor* quantization parameters:

$$E = \underbrace{(e_1^t, \dots, e_{m-1}^t)}_{\text{text scale } s_t} \underbrace{(e_m^v, \dots, e_n^v)}_{\text{visual scale } s_v} \underbrace{(e_{n+1}^t, \dots, e_L^t)}_{\text{text scale } s_t}. \quad (2)$$

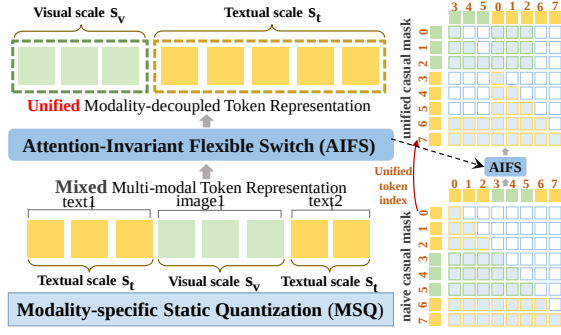


Figure 2: Overview of Modality-Specific Static Quantization (MSQ) and Attention-Invariant Flexible Switching (AIFS). AIFS reorders tokens so that all visual tokens appear first, then the textual tokens, while adjusting the causal mask to preserve the original model logic.

The scaling factors s_v and s_t correspond to visual and textual tokens, respectively. Our innovation lies in performing *single-shot calibration* of these factors before inference, thereby eliminating per-token computational overhead. This per-tensor static quantization strategy provides three distinct advantages: (1) complete avoidance of dynamic scale updates, (2) hardware-friendly implementation benefiting from simplified quantization kernels, and (3) prevention of distribution saturation where high-magnitude visual activations could overwhelm the narrower dynamic range of textual tokens.

Attention-Invariant Flexible Switching (AIFS). Despite the benefits of MSQ, it complicates data handling if textual and visual tokens remain interleaved in E . Naïve slicing and concatenating at runtime can increase memory traffic and reduce efficiency for GEMM-based layers like QK and FC. To overcome this, we propose *Attention-Invariant Flexible Switching (AIFS)*. Figure 2 illustrates the idea: we *reorder* the sequence so that all visual tokens appear first, followed by textual tokens. We then alter the causal mask to preserve the original auto-regressive relationships. Causal attention [42] ensures each token can only attend to earlier tokens. Formally, for a naive sequence E , the attention matrix is

$$A = \text{Softmax}\left(\frac{QK^T}{\sqrt{D}} + M_{i,j}\right), \quad M_{i,j} = \begin{cases} 0, & \text{if } j \leq i, \\ -\infty, & \text{otherwise.} \end{cases} \quad (3)$$

With AIFS mechanism, we can reorder E and obtain an unified causal mask $M_{i,j}^u$, the new $E^u = \{e_m^v, \dots, e_n^v, e_1^t, \dots, e_{m-1}^t, \dots, e_{n+1}^t, \dots, e_L^t\}$, and $M_{i,j}^u$ could be formulated as:

$$M_{i,j}^u = \begin{cases} 0 & \text{if one of the following conditions is met:} \\ & (i \leq (n-m), j \leq i \text{ or } (n-m) < j \leq n) \\ & \text{or } ((n-m) < i \leq n, (n-m) < j \leq i) \\ & \text{or } (i > n, j \leq i) \\ -\infty & \text{otherwise} \end{cases} \quad (4)$$

The position embeddings are also shifted accordingly (see Appendix A.2), preserving *numerical equivalence of attention scores* with the original sequence. This maintains auto-regressive consistency while enabling streamlined memory access: all visual/textual tokens can now be processed through respective static-scale multiplications.

Seamless Integration with FlashAttention. MSQ and AIFS, which optimizes MLLMs through optimizing on input token efficiency, while FlashAttention [14] accelerates attention computation via reduced memory access and tiling. These approaches are conceptually orthogonal, suggesting seamless integration with additive benefits without inherent conflicts. Specifically, at the beginning of inference, MSQ+AIFS reorders the intermixed multimodal tokens and passes the original visual modality start/end indices (m and n) to FlashAttention to ensure that irrelevant positions are correctly masked. Detailed results are in Table 6.

Efficiency and Practical Benefits. Our experiments in Section 4 show that MSQ + AIFS delivers three main advantages: ① *High Compatibility and Strong Theoretical Equivalence*: The causal-mask transformation ensures that the model’s output is numerically the same as if tokens were not reordered. This reordering can be integrated into existing MLLM implementations without major code changes. ② *Reduced Inference Latency*: We replace per-token dynamic quantization with a static per-modality approach. This removes the need to recalculate scales per token, cutting runtime overhead and boosting throughput. ③ *Enhanced Memory Efficiency*: Sequential processing of visual (via s_v) and textual (s_t) tokens with dedicated scale factors minimizes memory fragmentation, yielding up to 24.7% speedups and 152.9% memory efficiency gains (Table 8) through eliminated padding/slicing operations.

In summary, *MSQ + AIFS* offers a straightforward and effective way to handle the unique challenges of multi-modal token distributions. It is valuable on edge devices or other constrained platforms where dynamic quantization is hard to implement at scale, yet per-tensor static quantization is hardware-friendly and fast [38].

3.2 FHT-induced Weight Outlier Mitigation: Rotation Magnitude Suppression

QuIP [8] and Quip# [41] formalized *incoherence* to measure the difficulty of quantization. A lower incoherence indicates a simpler quantization scenario. For a weight matrix $W \in \mathbb{R}^{m \times n}$ with singular vectors e_i, e_j and Frobenius norm $\|W\|_F$, the *incoherence coefficient* μ is defined as the minimal constant satisfying:

$$|W_{ij}| = |e_i^T W e_j| \leq \mu \frac{\|W\|_F}{\sqrt{mn}}, \quad (5)$$

Generally, the smaller μ is, the easier it is to quantize W . They also showed that applying a Hadamard transform to both weights and activations can effectively reduce μ . However, Multi-modal LLMs (MLLMs) combine an LLM component with a *visual encoder*, which often relies on LayerNorm. Quarot’s partial online transform cannot be applied directly to all such norms. Inspired by SliceGPT [4], we convert visual encoders’ LayerNorms into RMSNorms. More details can be found in the Appendix A.14. This change makes Quarot-like Hadamard transforms applicable to MLLMs. Yet, as Table 3 demonstrates, Quarot still underperforms on many MLLM tasks. Quip# [41] also proved that *random Hadamard transforms* (RHT) can reduce incoherence for Hessians and weights, but they did not analyze *online* (fast) Hadamard transforms (FHT). In Quarot, FHT is crucial for low-bit quantization (e.g., 4-bit) as shown in Appendix A.9. We investigate why FHT can yield fresh outliers in MLLMs and propose a method to suppress them.

Channel-Mean Outliers in Online FHT. Following (Eq. 5), let $W_{\ell_2} \in \mathbb{R}^{n \times m}$ be a weight matrix. A Hadamard matrix $H \in$

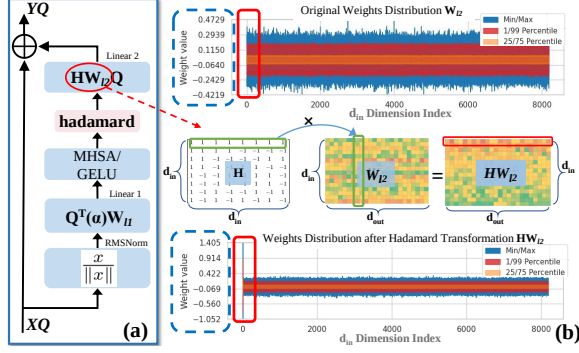


Figure 3: (a) The pipeline of Quarot, showing offline and partially online Fast Hadamard transforms. (b) Weight matrix where applying online FHT produces outliers in the first row.

$\{-1, 1\}^{n \times n}$ with orthonormal rows preserves this norm, i.e., $\|H W_{l2}\|_F = \|W_{l2}\|_F$. Suppose μ_{l2} is the incoherence coefficient for W_{l2} , satisfying $\max(|W_{l2}|) = \mu_{l2} \frac{\|W_{l2}\|_F}{\sqrt{mn}}$. Let HW_{l2} be the transformed weight, and let its incoherence coefficient be μ_{Hl2} . Then we have:

$$\frac{\mu_{Hl2}}{\mu_{l2}} = \frac{\max|H W_{l2}|}{\max|W_{l2}|}. \quad (6)$$

For many Hadamard transforms, the first row (and first column) contain identical $\frac{1}{\sqrt{n}}$ entries, while other rows sum to zero. Hence, the first channel after transformation is

$$(HW_{l2})_{0j} = \sqrt{n} \text{mean}(w_{:,j}). \quad (7)$$

If the mean is large, the first element in that row can exceed the original maximum and thus raise μ_{Hl2} . Concretely, when

$$\sqrt{n} \text{mean}(w_{:,j}) > \max_i (w_{ij}), \quad (8)$$

a new *channel-mean outlier* appears in the first row. Figure 3(b) shows such an occurrence in an MLLM. This issue arises especially in Quarot’s *online* (partial) FHT step, which applies Hadamard rotations per forward pass rather than strictly offline.

Table 2: Compliance Ratio with Eq. 9 Across Model Components

Model	Part	All Blocks Number	Meet Eq. 9 Number	Ratio
Internvl2-8B	Visual	24	24	100%
	LLM	32	4	12.5%
Qwenvl-9.6B	Visual	48	48	100%
	LLM	32	13	40.6%
MiniCPM-V-2.6-8B	Visual	27	27	100%
	LLM	28	3	10.7%
GLM-4V-9B	Visual	63	63	100%
	LLM	40	6	15.0%
Qwen2vl-7B	Visual	32	32	100%
	LLM	28	1	3.5%

Prevalence of Eq. 8 Compliance. We conduct systematic block-level verification across five state-of-the-art MLLMs (InternVL2-8B, QwenVL-9.6B, MiniCPM-V-2.6-8B, GLM-4V-9B, and Qwen2VL-7B) to quantify the occurrence of Fast Hadamard Transform (FHT)-induced weight outliers. As evidenced in table 2, our analysis reveals two critical findings: (1) **Universal Presence in Vision Encoders:**

All visual *downproj* layers (100%) exhibit FHT-induced weight outliers that satisfy Eq. 8, as conceptually illustrated in Fig. 3(b). (2) **Variable Manifestation in LLMs:** The phenomenon appears in only 3%–40% of LLM *downproj* layers across different architectures. This empirical validation confirms above theoretical soundness and the critical need for specialized handling of FHT-generated weight outliers, particularly in vision components where their universal presence fundamentally impacts model quantization.

Rotation Mitigation Scheme (RMS). To this end, we propose RMS to handle these new outliers with minimal overhead. We first identify whether a channel meets (Eq. 8). If it does, we: ① **Split** the problematic channel from the main GEMM kernel and process it using a separate GEMV. ② **Zero out** that row in HW_{l2} so that the main kernel does not double-count it. ③ **Add** the partial output (from the split GEMV) back to the main path before the activation. Figure 4 shows the workflow. This targeted split ensures large-mean channels do not trigger extreme first-row values during the per-forward-pass FHT. The added cost is small, since only channels meeting Eq. 8 require this procedure. Detailed RMS algorithm in Appendix A.7 outlines how RMS integrates with Quarot’s FHT. Overall, RMS addresses a key shortcoming of online Hadamard transforms by suppressing large-mean channels.

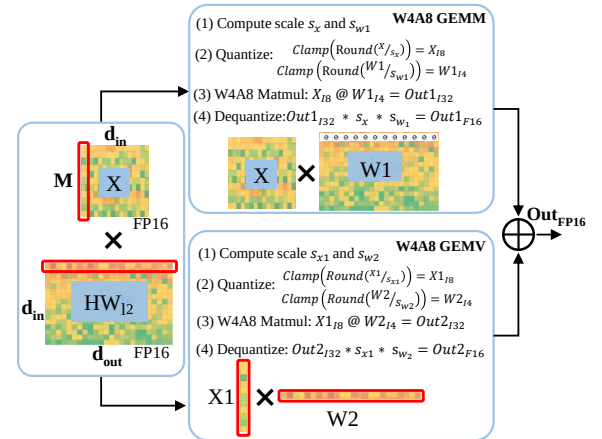


Figure 4: An overview of our proposed RMS. We separate outlier-prone channels into a dedicated GEMV path, zero their row in the main kernel, and then merge the results.

Why Not Subtract Channel Means Directly? A straightforward idea is to separate out each channel’s mean before applying H and then re-inject it afterward. However, this leads to two major issues: 1. The separated means are still subject to Hadamard rotation, creating a new linear transformation where the first channel again becomes large. 2. Splitting and then re-injecting the means effectively doubles the linear operations, significantly increasing the computational cost. In contrast, our RMS approach modifies only the row triggering the outlier condition and does not require additional linear layers, thereby offering a more efficient and effective resolution.

4 Experiments

Table 3: Comprehensive quantization results of different MLLMs across various multimodal reasoning datasets.

MLLMs	Method	Bits Setting		T.VQA↑	D.VQA↑	OCRB.↑	MME↑
		Visual	LLM				
InternVL2-8B	-	BF16	BF16	77.65	90.97	794	2209
	RTN			52.02	59.04	542	1528
	SQ	W8A8	W4A8	59.88	59.75	544	1540
	Quarot			73.34	84.07	715	2067
	MQuant			77.49	90.27	785	2156
	RTN			40.06	31.58	302	1482
	SQ	W4A8	W4A8	46.48	31.21	310	1540
	Quarot			49.10	33.62	361	1941
Qwen-VL-Chat-9.6B	-	BF16	BF16	61.40	60.36	493	1834
	RTN			0.45	0.03	189	625
	SQ	W8A8	W4A8	7.45	7.70	160	797
	Quarot			45.32	42.44	286	940
	MQuant			61.16	59.31	483	1691
	RTN			1.02	0.02	193	585
	SQ	W4A8	W4A8	8.59	4.28	188	921
	Quarot			46.77	37.35	289	1091
MiniCPM-V 2.6-8B	-	BF16	BF16	79.10	89.18	847	2248
	RTN			61.00	65.16	332	1300
	SQ	W8A8	W4A8	62.40	65.76	424	1510
	Quarot			73.71	80.04	736	1850
	MQuant			80.41	89.15	844	2244
	RTN			60.70	62.23	351	1404
	SQ	W4A8	W4A8	65.67	60.02	455	1491
	Quarot			68.96	79.63	685	1734
GLM-4V-9B	-	BF16	BF16	82.82	81.16	782	2153
	RTN			7.05	3.70	0.00	140
	SQ	W8A8	W4A8	9.05	4.10	0.00	148
	Quarot			82.00	80.17	782	2115
	MQuant			82.06	80.53	782	2164
	RTN			7.61	3.60	0.00	163
	SQ	W4A8	W4A8	9.85	4.40	0.00	188
	Quarot			64.16	45.52	516	2048
Qwen2-VL-7B	-	BF16	BF16	84.43	93.87	842	2319
	RTN			33.92	52.61	442	1298
	SQ	W8A8	W4A8	49.11	53.97	444	1500
	Quarot			79.36	89.57	754	2045
	MQuant			84.43	93.61	830	2269
	RTN			40.20	38.82	422	1082
	SQ	W4A8	W4A8	46.25	52.36	411	1535
	Quarot			71.44	83.96	670	1911
Qwen2-VL-72B	-	BF16	BF16	85.48	95.95	883	2479
	RTN			37.21	40.94	426	1444
	SQ	W8A8	W4A8	50.33	55.41	480	1601
	Quarot			80.03	91.21	781	2299
	MQuant			85.48	95.90	880	2469
	RTN			28.21	25.94	426	1137
	SQ	W4A8	W4A8	47.11	54.95	413	1586
	Quarot			71.86	86.11	701	2264
GLM-4V-9B	-	BF16	BF16	81.58	79.67	754	2120
	RTN			7.05	3.70	0.00	140
	SQ	W8A8	W4A8	9.05	4.10	0.00	148
	Quarot			82.00	80.17	782	2115
	MQuant			82.06	80.53	782	2164
	RTN			7.61	3.60	0.00	163
	SQ	W4A8	W4A8	9.85	4.40	0.00	188
	Quarot			64.16	45.52	516	2048

Models and Datasets. We evaluate our *MQuant* on five MLLMs: InternVL2-8B [11], Qwen-VL-Chat-9.6B [6], MiniCPM-V 2.6-8B [50], Qwen2-VL-7B [44], and GLM-4V-9B [19]. Evaluations are conducted on four benchmarks covering OCR and general question answering: TextVQA [36], DocVQA [32], OCRBench [31], and MME [18], which assesses perception and cognition across 14 tasks.

Baselines and Implementation Details. We test W8A8 and W4A8 quantization settings for both visual encoders and LLMs,

comparing RTN, SmoothQuant [47], and Quarot [5]. Notably, we apply static per-tensor activation quantization for both components, unlike the dynamic per-token quantization typically used in existing MLLMs. The calibration dataset consists of 256 randomly selected samples from the corresponding benchmark training sets [31, 32, 36]. The batch size for latency measurement is 1.

4.1 Overall Results

As shown in Table 3, *MQuant* can apply to the quantization of various MLLMs, and demonstrates significant improvements over several representative quantization methods. In W8A8 setting, *MQuant* achieves near-lossless performance to the FP models across diverse datasets and different model scales (7-72B). Notably, even in the more challenging W4A8 setting, *MQuant* maintains comparable performance with FP models, while other advanced quantization methods exhibit significant performance degradation. These results indicate that our *MQuant* provide a general and effective PTQ solution with strong compatibility for maintaining high accuracy in MLLMs under various bits settings across diverse multi-modal reasoning tasks. This suggests *MQuant* maintains robustness (e.g., stability to typical input or models changes) in these frameworks.

4.2 Ablation Study

In this section, we select Qwen2-VL-7B [44], one of the most powerful open-source MLLMs, to validate the effectiveness of our designs.

Table 4: Ablation study of proposed designs on Qwen2-VL-7B [44] with W4A8 setting.

	Methods			T.VQA↑	D.VQA↑	OCRB.↑	MME↑	Lat (ms) ↓
	Static	AIFS + MSQ	RMS					
				84.43	93.87	842	2319	6523
✓		✓	✗	71.44	83.96	670	1911	5479
✓		✓	✗	78.95	87.55	721	2095	5484
✓		✓	✓	84.32	93.58	824	2255	5471

Ablation Study of Proposed Quantization Designs. We perform ablations on Qwen2-VL-7B (Table 4) to isolate the contributions of each quantization component. Beginning with a baseline that applies GPTQ + Hadamard transformations to both the LLM and vision parts with per-tensor static quantization, we progressively introduce **MSQ+AIFS** for the LLM. This significantly boosts accuracy and latency, highlighting their effectiveness on multi-modal input tokens. Next, incorporating **RMS** to suppress newly arising outliers achieves near-floating-point performance, underscoring the overall robustness and efficiency of our pipeline.

Table 5: Ablation study about calibration dataset size.

Calib size	T.VQA↑	D.VQA↑	OCRB.↑	MME↑
128	84.28	93.50	820	2243
256	84.32	93.58	824	2255
512	84.32	93.57	823	2254

Selection of Calibration Dataset Size. Following [5, 47], we selected a sample subset to get the statistics of activations during the calibration process. This ensures the calibration data reflects the distribution in each dataset. To assess sensitivity to calibration dataset size of *MQuant*, we conducted experiments on Qwen2-VL-7B, including [128, 256, 512] samples. The table shows the consistency of *MQuant* in W4A8 with lossless accuracy. These results, revised

in supplementary, demonstrate that MQuant is largely insensitive to calibration dataset size. We believe this robustness enhances MQuant’s practical applicability. Therefore, we select 256 randomly selected samples from the corresponding benchmark training set.

Table 6: Latency combining AIFS with FlashAttention from 100 to 8100 tokens ($Q/K/V$ shape is (1, 28, num-tokens, 128))

Tokens Number	100	900	2500	4900	8100
FlashAttn (ms)	27.3	33.6	35.5	165.9	271.6
AIFS+FlashAttn (ms)	27.3	34.1	36.4	167.5	272.1

Integrate AIFS with Flash Attention. As evidenced in Table 6, we integrated the causal attention mechanism of AIFS into FlashAttention and tested it on an RTX 6000, observing that its speed is nearly identical to that of standard causal attention. The results demonstrate seamless integration MBQ+AIFS with Flash Attention, exhibiting negligible latency overhead ($<0.3\%$ across all tested token lengths). The minimal performance gap (maximum 1.6 ms) stems from two key design choices: (1) AIFS’s token reordering operates exclusively in the preprocessing phase without interfering with core attention computation, and (2) our metadata propagation mechanism (visual token indices m/n) preserves Flash Attention’s native memory access patterns. Notably, the 0.96% latency increase at 4,900 tokens validates strong scalability for high-resolution inputs. This highlight the hardware compliance advantages of MBQ and AIFS designs: CUDA-aligned token layouts enable binary compatibility with FlashAttention kernels, which is particularly effective for real-time processing of high-resolution multimodal inputs.

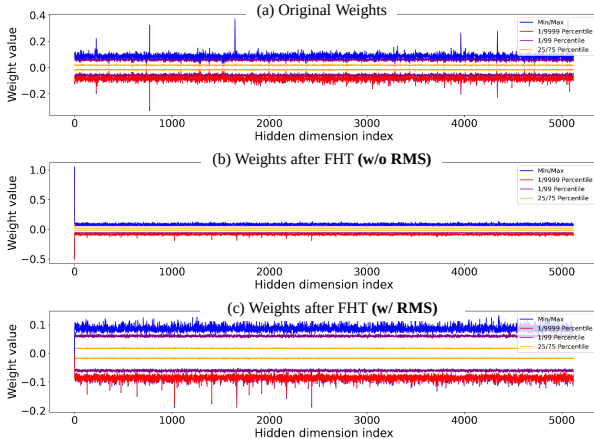


Figure 5: Illustration the weight distributions for the down-proj layer (block 21) in Qwen2VL-7B’s visual encoder under three conditions: (a) original weight, (b) weights after FHT, and (c) weight after FHT with our RMS.

Weight Distribution with Rotation Magnitude Suppression (RMS). As shown in Fig. 5, we visualizes the magnitude of weights outliers without (w/o) and with (w/) RMS. The weight distribution reveals that FHT amplification increases weight magnitudes, while RMS effectively reduces these magnitudes by a factor of 1.0 to 0.1, significantly improving quantization stability (Other distribution are in Appendix). This is also consistent with our ablation results, shown that RMS significantly enhances performance for both MLLMs (Table 4). Taking Qwen2-VL as example, the vision encoder’s down-projection layer (3584×1280 , $C_{in} \times C_{out}$) and LLM’s

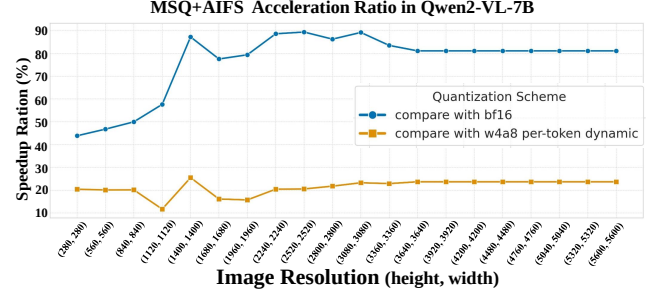


Figure 6: The Speedup of AIFS+MSQ on Qwen2-VL-7B.

corresponding layer (18944×3584) demonstrate the efficiency of our approach. The RMS implementation converts a W4A8 GEMM operation into a combined W4A8 GEMM + W4A8 GEMV computation, where the GEMV is selectively applied only to affected channels. This introduces negligible overhead, precisely $1/3584$ and $1/18944$ of the total computation for vision and LLMs parts.

Accuracy & Speedup of AIFS. While MSQ addresses the modality discrepancy, AIFS further reorders visual and textual tokens into a unified sequence to enable efficient per-tensor static quantization of activations. Table 7 shows that MSQ+AIFS not only achieves the same speedup as naïve per-tensor static quantization but also maintains near-floating-point accuracy across linear layers. Figure 6 further illustrates that AIFS yields speedups of 20%–80% as resolution increases, corroborating that rearranging tokens avoids the high overhead of dynamic per-token quantization.

4.3 Latency and Memory Analysis

We evaluate MQuant’s performance across three key metrics: (1) inference speedup, (2) memory efficiency (tested over image resolutions from 280^2 - 5600^2), and (3) decoding acceleration. All experiments employ the established “text-image-text” input format in [15] with a fixed 15 textual tokens unless explicitly varied.

Inference Speedup and Memory Savings. ① **Overall Speedup** As shown in Table 8, MQuant surpasses BF16 and AWQ (W4-only) across all image resolutions, achieving up to **24.76%** speedup over PyTorch at 840×840 . Even at higher resolutions (e.g., 5600^2), MQuant maintains a **12.19%** latency improvement, demonstrating scalability. ② **Memory Efficiency.** Our method provides memory savings over both BF16 and AWQ, with reductions exceeding **100%** compared to PyTorch (e.g., **152.92%** at 840^2). These benefits primarily arise from (1) eliminating the overhead of token-wise scale computation, and (2) converting mixed tokens into modality-decoupled tokens, avoiding slicing and concatenation when image resolutions become large. ③ **Decoding Acceleration.** We also measure decode time for generating 2,000 tokens with a custom W4A8 GEMV kernel (Table 10). Compared to per-token dynamic quantization, our AIFS+MSQ framework gains **23%** speedup in the prefill stage and **100%** speedup in decode stage. By shifting on-line per-token dynamic quantization to an offline per-tensor static approach, we greatly reduce inference overhead, especially for long-sequence tasks. Moreover, since visual tokens are generally pricier than textual ones [15], these improvements translate to notable real-world cost reductions (e.g., $\approx 30\%$ for OpenAI token pricing).

Acceleration for Multi-batch and Multi-turn Inference. ① **For multi-batch scenarios,** we use 2240×2240 resolution images

Table 7: The accuracy and speedup of our MSQ and AIFS scheme on the linear layer during prefill stage. The input sequence is structured as "text-image-text-image-text-image-text" with an image resolution of 2240×2240 and 50 textual tokens. Latency were tested on an NVIDIA RTX 6000 Ada Generation.

Activation	Weight	T.VQA↑	D.VQA↑	OCR.B.↑	MME↑	Latency ↓ (s)	Speedup ↑
BF16	BF16	84.43	93.87	842	2319	1.690	-
BF16	W4-g128(AWQ)	83.93 (-0.50)	93.13 (-0.74)	828 (-14)	2252 (-67)	2.057 (+0.367)	-17.8%
A8-per-token dyn	W4-per-channel sta	84.32 (-0.11)	93.61 (-0.26)	830 (-12)	2269 (-50)	1.253 (-0.437)	+34.9%
A8-per-tensor sta		40.20 (-44.12)	38.82 (-54.79)	422 (-408)	1082 (-1187)	1.016 (-0.674)	+66.3%
A8-MSQ		84.32 (-0.11)	93.61 (-0.26)	830 (-12)	2269 (-50)	1.085 (-0.605)	+55.8%
A8-MSQ+AIFS		84.32 (-0.11)	93.61 (-0.26)	830 (-12)	2269 (-50)	1.017 (-0.673)	+66.2%

Table 8: Comparison of latency and memory saving with Pytorch (BF16), AWQ (W4-only) and MQuant (W4A8) on Qwen2-VL-7B. ↓ means lower values are better, ↑ means larger values are better.

Image size		Latency (s)				Memory (G)				
H × W	Pytorch	AWQ↓	Speedup↑	MQuant↓	Speedup↑	Pytorch	AWQ↓	Improve↑	MQuant↓	Improve↑
840 ²	0.261	0.304 (+0.043)	-14.14%	0.210 (-0.051)	+24.76%	16.45	7.45 (-9.00)	+120.67%	6.50 (-9.95)	+152.92%
1960 ²	1.369	1.598 (+0.229)	-14.26%	1.112 (-0.257)	+16.63%	17.82	8.82 (-9.00)	+100.00%	7.85 (-9.97)	+119.93%
3080 ²	5.208	5.872 (+0.664)	-11.27%	4.488 (-0.720)	+16.02%	20.58	11.58 (-9.00)	+77.60%	10.61 (-9.97)	+96.45%
5600 ²	8.380	9.393 (+1.013)	-10.78%	7.469 (-0.911)	+12.19%	22.22	13.22 (-9.00)	+57.54%	12.25 (-9.97)	+61.65%

Table 9: Multi-Batch speedup comparison of MSQ + AIFS on W4A8 setting. Each row shows the cumulative total of text tokens, images, and textual responses for multi-turn inference.

Batch	Text	Config (Text+Image+Text)			Prefill (s)			Decode (s)			All (s)		
		Image	Text	Text	bfp16	MQuant	Improve↑	bfp16	MQuant	Improve↑	bfp16	MQuant	Improve↑
1	10	2240×2240	50	50	2.54	1.93	+31.6%	18.01	12.89	+39.7%	20.55	14.82	+38.7%
2	10/10	2240×2240 / 2240×2240	50/100	50/100	5.42	4.15	+30.6%	37.82	31.56	+19.8%	43.24	35.71	+21.1%
3	10/10/10	2240×2240 / 2240×2240 / 2240×2240	50/100/150	50/100/150	8.24	6.42	+28.3%	48.03	40.35	+19.0%	56.27	46.77	+20.3%
4	10/10/10/10	2240×2240 / 2240×2240 / 2240×2240 / 2240×2240	50/100/150/200	50/100/150/200	11.17	8.67	+28.9%	59.09	49.92	+8.4%	70.26	58.59	+20.0%

Table 10: Speedup of MSQ + AIFS on W4A8 setting.

Stage	BF16	Per-token Dyn.	Ours	Ours + GEMV	Speedup
Prefill	1690	1253	1017	-	+23%
Decode	17.5	16.4	13.06	8.2	+100%

Table 11: Latency comparison under multi-turns setting.

Turns	Config in one Turn			All(s)		Improve ↑
	Text	Image	Text	bfp16	Ours	
1	10	2240×2240	50	20.55	14.82	+38.7%
2	10	2240×2240	50	44.06	32.61	+35.1%
3	10	2240×2240	50	76.67	59.48	+28.9%

with "text-image-text" inputs, we evaluate batches containing 50-200 text tokens while generating 512 decoded tokens. Sequence alignment employs left-padding (pad_token_id) with proper masking (see Appendix A.4). AIFS maintains full multi-batch compatibility without overhead, enabling consistent 20% faster inference versus FP baselines across batch sizes 1-4 (Table 9). **2. For multi-turn dialogue**, under identical resolution and input format (50 text tokens per turn + 512 decoded tokens), MQuant achieves 38.7% faster end-to-end inference than BF16 for 1-3 turns (Table 11) through optimized key-value caches and position IDs across turns.

Weight-only Quantization. We evaluate our weight-only quantization using GPTQ [17] and AWQ [29] under consistent settings (group size=128), quantizing only the LLM while keeping the visual encoder in BF16. As shown in Table 12, our W4 LLM quantization achieves near-lossless accuracy comparable to existing methods. When extended to W4 quantization, MQuant maintains comparable performance with the BF16 baseline. The W4A8 quantization

Table 12: Comparison under weight-only settings on Qwen2-VL-7B [44]. † means re-implementation based on the official weight-only setting with a group size of 128.

Method	Bits Setting		T.VQA ↑	D.VQA ↑	OCR.B.↑	MME↑
	Visual	LLM				
-	BF16	BF16	84.43	93.87	842	2319
GPTQ (g128)†	BF16	W8	84.33	93.97	842	2313
GPTQ (g128)†	BF16	W4	84.18	93.25	831	2285
AWQ (g128)†	BF16	W4	83.93	93.13	828	2252
MQuant (g128)	BF16	W4	84.55	93.18	832	2304
MQuant (g128)	W4	W4	84.70	93.57	828	2292
MQuant	W4A8	W4A8	84.32	93.58	824	2255

similarly shows competitive or superior results compared to weight-only approaches. These experiments validate MQuant’s effectiveness across: (1) weight-only vs. weight-activation schemes, and (2) partial vs. full-model quantization scenarios.

5 Conclusion

In this paper, we propose *MQuant*, an accurate and efficient PTQ framework specifically designed for MLLMs. Our approach addresses the challenges of MLLMs by applying **Modality-Specific Static Quantization (MSQ)** to handle distribution mismatches between visual and text tokens, and introducing an **Attention-Invariant Flexible Switching (AIFS)** mechanism to reduce TTFT with static per-tensor scaling. Furthermore, we reveal that online Hadamard rotations can lead to weight outliers, and propose **Rotation Magnitude Suppression (RMS)** to mitigate this issue. Extensive experiments on 5 mainstream MLLMs demonstrate that *MQuant* achieves near-floating-point accuracy under the W4A8 setting, highlighting its potential to advance MLLMs quantization and practical deployment in resource-constrained environments.

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